TBWorldview

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Formatted for double-sided printing. Each major section starts on a recto (odd) page so the printout can be filed into a binder. Blank pages have been inserted where a section ended on an odd page.

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Original ToolBook Directory Contents

Files that belong in the Nida ATT ToolBook installation

Source of truth

The file **toolbook.lst** is the runtime load list. It names every book, DLL, and executable the Nida shell expects. **version.ini** reports 03.001.0006 dated 04-18-05. Modern course-menu rebuilds (COUR1.EXE, courses_menu.exe, and their MinGW runtimes) are not in this list and were not part of the original Nida ToolBook directory.

Nida system books and support

NC81BASE.SBK, NC81RECS.SBK, NC81TEST.SBK, NC81TRNR.SBK, NC81BANK.SBK
NC81INT.DLL, NC81REC.DLL
NIDA81.TBK
nida.ttf (font)
CAILT.DAT, STDAPPS.LST, version.ini

Core ToolBook 8 binaries

TB80RTM.EXE (engine), TB80RUN.EXE, TB80MAIN.EXE, Tbload.exe
TB80BAS.DLL, TB80BMP.DLL, TB80CMP.DLL, TB80UTL.DLL
TB80W16.DLL, TB80W32.DLL, TB80ENU1.DLL, TB80ENU2.DLL
TB80OLE.DLL, TB80VBX.DLL, TB80OCX.DLL, TB80ACX.DLL
TB80FLT.DLL, TB80LNL.DLL, TB80MM.DLL, TB80NET.EXE
TB80RCR.DLL, TB80RMT.DLL, TB80JPG.DLL, TB80F32L.DLL
TBCBT.DLL, TBDB3.DLL, TBDLG.DLL, TBDOS.DLL
TBFILE32.DLL, TBPDX.DLL, TBREG.DLL, TBWIN.DLL

System books

TB80R.SBK, TB80ANM.SBK, TB80HYP.SBK

Not part of the original directory

COUR1.EXE / COUR1.BAS, COURMENU.*, COURMENV.*, thecourse*.bas
courses_menu.exe / courses_menu.bas and the CSV exports
libgcc_s_seh-1.dll, libstdc++-6.dll, libwinpthread-1.dll (MinGW runtimes for the modern menu)
courses/ subdirectory, runtime2/, courses.xml (menu export)
zlib1.dll (not listed in toolbook.lst)

ToolBook 8 Reverse Engineering

Overview & Status Report

October 2026 — Nida ATT Master_3 Courses

1. What These Books Are

The course files are classic Asymetrix/Click2learn ToolBook books (ToolBook II 8.1 shells, book version 01 00 00 08). Each **.tbk** begins with a short PE stub (0xC00 bytes, no real code) followed by the book body. The magic **ZZZTBKBOOKLOC** and the value **0x22222222** mark the start of the real book data. The opener is **tbload.exe** via **MINSTB32.dll**.

The runtime engine that actually interprets the book is the 16-bit NE binary **TB80RTM.EXE** (unpacked from its SLR Optloader wrapper). Supporting DLLs (**TB80BAS**, **TB80CMP**, **TB80BMP**, **TB80UTL**, **TB80W16**, and others) supply object types, painting, OpenScript compilation/expansion, bitmaps, and windowing.

The courses themselves are Navy electronics CAI. **courses.xml** is a menu export that lists 0_MASTER3 plus twenty rating tracks. On disk there are hundreds of lesson **.tbk** files. Work has concentrated on **9e954d01.tbk** (Logic Families) plus the shared shell **nida81.tbk** and the system books.

2. On-Disk Structure (Understood)

Page objects are the only records that begin with the marker **80 65 00 05 00 02 02**. Inside each page the widgets are nested chunks that begin **14 00 00**. A typical property block starts **14 00 00 01**. Field text is stored as a Text property (length-prefixed, often with a leading CRLF).

Standard property IDs (bounds = 0x4004, text = 0x402e, script = 0x4023, vertices = 0x4034, etc.) were recovered from NE resource type 0x2d1 / id 0x2d0 in TB80BAS.

Object-type names (Page, Button, Field, Picture, Polygon, Arc, Curve, Rectangle, etc.) live in a classification table inside TB80UTL.DLL and are also present as resources in the system .sbk files.

3. What Modern Readers Already Do

Several generations of clean-room readers exist:

• Early HTML/XML viewers (viewer, viewer2, viewer3) that walk the page marker, extract text and quoted jumps.
• Maybe2 / Maybe4 lessonshells: JSON with pages, Next/Back, quiz choices, and decoded script.
• runtime/ and runtime_cpp/: Python and C++ packages that parse a book and emit HTML or drive a simple player.
• Brickwall / Foolbook: self-contained GUI readers that show page lists, field text, and quoted jumps.

All of these can already display the textual content, navigation, and quizzes of books that resemble Logic Families. They do not yet paint arbitrary polygons, arcs, curves, or placed bitmaps for the full corpus.

4. Function Naming Status

Pattern-based naming across decompiled modules has identified ~1,918 nameable functions (~17.5% of analyzed). Manual confirmation has produced over 100 named functions, concentrated in paint, scale, property, memory, and string helpers.

See **Modules/** subdirectories for per-module confirmed names and nameable lists. See **03_Naming_Progress.pdf** for the full summary.

5. Open Work

• Bounds extractor (record-to-object+0xF) still missing from decompiles.
• Full OpenScript execution and scoring.
• Complete vector painting (arcs, curves, computed brackets) and bitmap placement.
• NE host that fully opens a book under emulation.

Document Map

This package is organized for printing:

Main directory — overarching documentation (this overview, architecture, naming progress).
Modules/tb80xxx_knownnames/ — confirmed function names and nameable lists per DLL.
BookFormat/ — on-disk record layouts, polygon/arc geometry, property IDs.
OpenScript/ — bytecode decoder notes and extracted handlers.
SoftICE/ — dynamic analysis notes and live object layouts.
Runtimes/ — clean-room reader status.

ToolBook 8 Architecture

Runtime flow, modules, and object model

High-Level Architecture

ToolBook is an interpreted, object-oriented, event-driven system built around a book metaphor. A .tbk file contains a PE stub, ZZZTBKBOOKLOC magic, and a JBO body of page records. TB80RTM.EXE is the engine: book loader, page table, message dispatch, and window classes (TBKMain, TBKClient, TBKChildWindow, SchedulerWndTB80). Supporting DLLs provide objects, painting, OpenScript, bitmaps, and type tables.

Key Modules (16-bit NE unless noted)

Module	Role
TB80RTM.EXE	Main engine, book open, windowing, message loop
TB80BAS.DLL	Objects, properties, painters, scale twips, type dispatch
TB80CMP.DLL	OpenScript bytecode, keyword pool, expand/emit
TB80BMP.DLL	Bitmaps, DIB, palette, graphic FX, LZ77
TB80UTL.DLL	Object type table (Page, Polygon, Field, …), helpers
TB80W16.DLL	Thunk layer, graphic IDs, script editor commands
nida81.tbk + .sbk	Shared shell, system books (base, recs, test, trnr, tb80r)

Book Open Path (partial)

tbload finds ZZZTBKBOOKLOC → JBO signature checked by virtual method → page structures allocated → CDBSENDCREATE with bounds → FUN_12c8_0850 creates object and writes bounds to object+0xF. The record-to-bounds extractor itself has not yet been located in the decompiles.

Painting Path

Live objects carry bounds at +0xF. Polygon painter (TB80BAS seg 0x82) reads count at +0x30 and point array at +0x32, scales twips via ScaleTwipsToScreen ((stored*96+720)/1440), then calls GDI!POLYGON. Arc painter (seg 0x7B) uses four twip pairs; rectangle painter reads bounds and calls GDI!RECTANGLE. Many shapes are computed from object bounds rather than stored point lists.

OpenScript

Handlers live in page records. Keyword tables are NE resources 0x77/0x78 in TB80CMP. Decoder recovers readable source (“go to page "P400"”, “send next”) including 0xC0–0xFF back-references from the compressed input stream.

Function Naming Progress

Pattern-based nameable + confirmed renames (2026-10-11)

Overall Coverage

Modules analyzed: 8 of 13. Unique functions (analyzed): ~10,991. Nameable by pattern: ~1,918 (17.5%). 30% target would be ~3,297; gap ~1,379. Confirmed (manual) names exceed 100 and are concentrated in paint, scale, property, memory, and string helpers.

Per-Module Nameable

Module	Unique	Nameable	%
TB80BAS	3,739	650	17.4%
TB80RTM	2,669	873	32.7%
TB80CMP	1,058	59	5.6%
TB80BMP	381	99	26.0%
TB80UTL	192	79	41.1%
TB80W16	330	57	17.3%
TB80LNL	805	98	12.2%

Confirmed Names by Category (selected)

Paint/Draw: PaintPolygon, PaintPolyline, PaintArc, PaintEllipse, PaintRoundRect, PaintRectangle, BitBlt helpers, CreateDibBitmap, Lz77Expand.
Scale/Property: ScaleTwipsToScreen, SetProperty, SetPointArray, SetupPaintRect, SetMultiProperty, ValidateProperty.
Memory/String: Alloc/Lock/StrCpy/StrLen/StrCat helpers across modules.
Compiler: LoadCompilerResource, ExpandTextCore, CompilerStr*.

Notes

RTM is high because it is the main engine with heavy GDI, message, and property dispatch. CMP is low because it is mostly compiler emit routines and resource tables. Confirmed names are the ones manually verified and present in the rename_*.py Ghidra scripts. See Modules/ subdirectories for full confirmed lists and nameable TSVs.

Book File Format

On-disk records, properties, and geometry

File Header

Each .tbk begins with a 0xC00-byte PE stub containing no executable code. The real book starts at the ZZZTBKBOOKLOC marker (file offset 0xC00 + magic 0x22222222). Book signature is typically 03 4A 42 4F 01 00 00 08 (JBO + version).

Page Records

Page objects are the only records that begin with the marker **80 65 00 05 00 02 02**. Name is a 32-byte field. Property block at +43 starts 14 00 00 01. Widgets are nested 14 00 00 chunks inside the page.

Property IDs (from TB80BAS resource)

ID	Name	Notes
0x4004	bounds	Object bounds (twips)
0x402e	text	Field text (length-prefixed, often leading CRLF)
0x4023	script	OpenScript bytecode
0x4034	vertices	Point array route
0x4050	point-array setter	Runtime only; stores count + points
0x4015	idNumber	Object ID

Polygon Geometry (solved)

Draw records are 14 00 00 subtype-0 chunks with +0x0c = 04 00. Point count is the word at +0x1a; x,y twip pairs start at +0x22. Fill color dword is at +4 (e.g. 0xC0C0C0 silver). Screen coordinates = (stored * 96 + 720) / 1440 (or stored/15 when DPI factors are 96). Confirmed on P430 pin tips and chip body. Individual pin positions use an index at +0x52.

Arc Geometry

Type 0x11. Four twip pairs at +0x25/+0x27, +0x29/+0x2B, +0x2D/+0x2F, +0x31/+0x33. Bounding box at +0x0F. Fill color at +0x18. Painter is TB80BAS segment 0x7B. ToolBook does not call GDI!PIE; pie wedges are computed via GDIELLIPSEPOINTFROMANGLE then filled.

Object Type Table

TB80UTL.DLL at file 0xDED6 contains the classification table: ComboBox, Stage, Picture, Hotword, RecordField, PaintObject, Curve, AngledLine, Pie, Arc, IrregularPolygon, Polygon, Line, RoundedRectangle, Ellipse, Group, Field, Button, Rectangle, then Page, Background, Book. Resource kinds after *MEMBLOCK* include cel, anim, ref, sharedscript, clip, font, menubar, palette, cursor, icon, bitmap.

Open Items

- Absolute bounds are not stored; position is relative to a parent.
- Many connecting lines and detail strokes are computed from object bounds by segment 0x5B rather than

stored.
• Curve type ID is 0x05C7; control structure for embedded point groups still incomplete.
• Record-to-bounds extractor (the code that reads a 14 00 00 chunk and writes object+0xF) has not been located.

OpenScript Decoder

Bytecode expansion and extracted handlers

Decoder Status

Keyword offset table is at byte 2 + count*4 in resource 0x77 (TB80CMP). Back-references 0xC0–0xFF copy from the compressed input stream (not the output). Decoder openscript_decode.py produces readable handlers.

Example Recovered Handlers (9e954d01.tbk)

• if page = false then go to page …
• send next
• go to page "P400"
• go to page "P310" of book "nida81.tbk"

All 76 handlers from Logic Families extract correctly. System books (nc81*) have no "to handle" OpenScript; nida81.tbk has ~100 buttonClick handlers in the same format.

Notes

Handlers are stored inside page records. The modern readers (Maybe2/4, Brickwall, Foolbook) already prefer decoded "go to page" / "send" / nextpage for navigation and quizzes. Full OpenScript execution (conditionals, scoring, system calls) is not yet implemented.

SoftICE Dynamic Notes

Live object layouts and paint traces

Window Classes (TB80RTM)

TBKMain, TBKClient, TBKChildWindow, TBKReserved_1, SchedulerWndTB80. A lesson click is delivered to TBKClient, not TBKChildWindow. BMSG does not see the 16-bit queue; BPX on the TBKClient proc does.

Polygon Painter Confirmation

TB80BAS segment 0x82 paints a polygon. GDI!POLYGON count is at object+0x30 (word), point array far pointer at object+0x32 (4 bytes each, stored twips). PaintPolygon = FUN_1408_0376. ScaleTwipsToScreen = FUN_1008_01dd (seg 2 offset 0x1DD). With DS:01A6/01A8 = 96, screen = stored/15. Breakpoint fires per polygon; leave it set while scrolling freezes SoftICE.

Arc and Rectangle

Arc painter: TB80BAS segment 0x7B, CALL GDI!ARC. Object far pointer at [BP+04]. Bounding box stored as twips at object+0x0F. Fill color at +0x18. Rectangle painter (seg 0x83) reads bounds at +0x0F and calls GDI!RECTANGLE. Chip bodies are the small rectangles.

Dump Caveats

Early /DUMP captures were SoftICE console logs, not target memory. With expert mode and careful addressing, real page and object dumps were obtained (ASYM_TpID, Page#, property blocks, live point arrays).

Clean-Room Runtimes

What already works and what remains

Existing Readers

• **viewer / viewer2 / viewer3**: HTML frame + XML/JSON books; page marker walk, text, quoted jumps.
• **Maybe2 / Maybe4**: lessonshell JSON with Next/Back, quiz choices, decoded script; path manager and quiz helpers.
• **runtime/ and runtime_cpp/**: Python and C++ packages that parse a book and emit HTML or drive a simple player.
• **Brickwall / Foolbook**: self-contained GUI (tbk_gui.py / python -m foolbook --gui) that serves raw .tbk or JSON shell.

What Works Today

Textual content, navigation, and quizzes for books that resemble Logic Families. System-book stubs supply path manager, bookmarking, and popup behavior. Polygon renderer for kind=0400 records (P430 chip example).

Still Open

• Full vector painting for arcs, curves, computed brackets, and placed bitmaps.
• OpenScript execution and scoring.
• Cross-book references and system-book integration beyond stubs.
• NE host that fully opens a book under emulation.

TB80BAS Known Names

Objects, properties, painters, twip scale

Confirmed Renames

Original	Name
FUN_1408_0376	PaintPolygon
FUN_13e8_00e6	PaintPolyline
FUN_10a0_0558	PaintArc
FUN_10a0_0458	PaintArcAlt
FUN_1020_03e4	PaintRectangle
FUN_1008_01dd	ScaleTwipsToScreen
FUN_12d0_03ec	SetProperty
FUN_1408_00d8	SetPointArray
FUN_12d0_086c	SetupPaintRect
FUN_1008_03f6	SetScriptProperty
FUN_1070_1774	ValidateProperty
FUN_1070_1b3c	CheckVertices
FUN_1230_2f1c	PaintEllipse
FUN_13d8_0066	PaintEllipseAlt
FUN_1230_2278	PaintRoundRect
FUN_1230_26f4	PaintRoundRectAlt
FUN_13e0_015a	PaintRoundRectAlt2
FUN_1230_2b5a	PaintArc2
FUN_10e0_208e	SetMultiProperty
FUN_10e8_33e8	SetMultiProperty2

Nameable Functions (pattern)

Full list is in bas_nameable.tsv (copied into this directory). Below is a sample of the first entries.

category	function	confirmed_name
GDI	FUN_1008_01dd	ScaleTwipsToScreen
GDI	FUN_1008_0290	—
GDI	FUN_1008_03f6	—
GDI	FUN_1008_0ac8	—
GDI	FUN_1010_3138	—
GDI	FUN_1010_43f2	—
GDI	FUN_1018_0000	—
GDI	FUN_1018_0cbe	—
GDI	FUN_1018_0e60	—

category	function	confirmed_name
GDI	FUN_1020_0176	—
GDI	FUN_1020_0b96	—
GDI	FUN_1020_16de	—
GDI	FUN_1028_11e2	—
GDI	FUN_1028_23ec	—
GDI	FUN_1028_37ae	—
GDI	FUN_1030_08ec	—
GDI	FUN_1030_0b80	—
GDI	FUN_1038_08a0	—
GDI	FUN_1038_08fc	—
GDI	FUN_1060_143e	—
GDI	FUN_1098_0514	—
GDI	FUN_1098_0818	—
GDI	FUN_1098_0b88	—
GDI	FUN_1098_0be0	—
GDI	FUN_1098_0c46	—
GDI	FUN_1098_0cda	—
GDI	FUN_1098_0d86	—
GDI	FUN_1098_0daa	—
GDI	FUN_1098_0dce	—
GDI	FUN_1098_0ec0	—
GDI	FUN_1098_0ee2	—
GDI	FUN_10a0_0018	—
GDI	FUN_10a0_00e6	—
GDI	FUN_10a0_0178	—
GDI	FUN_10a0_038a	—
GDI	FUN_10a0_03fa	—
GDI	FUN_10a0_0458	PaintArcAlt
GDI	FUN_10a0_04da	—
GDI	FUN_10a0_0558	PaintArc
GDI	FUN_10a0_061e	—

(showing first 40 of many; see TSV for complete list)

Notes

Rename script: rename_bas.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80BMP Known Names

Bitmaps, DIB, palette, LZ77 expand

Confirmed Renames

Original	Name
FUN_1078_05c2	CreateDibBitmap
FUN_1090_0558	CreateDibBitmapAlt
FUN_1098_0874	CreateDibBitmapAlt2
FUN_10a0_16ee	CreateDibBitmapAlt3
FUN_1048_0116	BitBltBmp
FUN_1048_1250	BitBltBmp2
FUN_1048_2c5e	BitBltBmp3
FUN_1080_0194	BitBltBmp4
FUN_1010_18a4	SelectPaletteBmp
FUN_1030_03e4	GetDcBmp
FUN_1080_0548	SelectPaletteBmp2
FUN_1000_0031	Lz77Expand
FUN_1000_0162	Lz77ExpandCaller
FUN_1000_0752	Lz77ExpandCaller2

Nameable Functions (pattern)

Full list is in bmp_nameable.tsv (copied into this directory). Below is a sample of the first entries.

category	function
File	FUN_1008_0432
File	FUN_1090_0512
File	FUN_1090_0558
File	FUN_10a0_0eee
File	FUN_10a0_0f38
File	FUN_10a0_16ee
File	FUN_10a0_194e
File	FUN_10a0_19ae
GDI	FUN_1008_0500
GDI	FUN_1008_07a8
GDI	FUN_1008_0898
GDI	FUN_1010_1e72
GDI	FUN_1010_1eb0
GDI	FUN_1010_21ce
GDI	FUN_1048_0116
GDI	FUN_1048_1090

category	function
GDI	FUN_1048_1132
GDI	FUN_1048_26de
GDI	FUN_1048_28a8
GDI	FUN_1048_2940
GDI	FUN_1068_005e
GDI	FUN_1068_00d2
GDI	FUN_1070_04f2
GDI	FUN_1070_0a48
GDI	FUN_1078_0a0a
GDI	FUN_1080_011c
GDI	FUN_1080_02c4
GDI	FUN_1080_0548
GDI	FUN_1080_057a
GDI	FUN_1080_0722
GDI	FUN_1080_094e
GDI	FUN_1088_05ee
GDI	FUN_1088_06d4
GDI	FUN_10b8_002c
Memory	FUN_1008_07a8
Memory	FUN_1008_0c20
Memory	FUN_1008_17d0
Memory	FUN_1008_1a70
Memory	FUN_1030_06ee
Memory	FUN_1038_011e

(showing first 40 of many; see TSV for complete list)

Notes

Rename script: rename_bmp.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80CMP Known Names

OpenScript compiler resources and expand

Confirmed Renames

Original	Name
FUN_1010_0ee6	LoadCompilerResource
FUN_1010_1390	LockCompilerResource
FUN_1038_0dea	LockCompilerResource2
FUN_1040_0970	ExpandTextCore
FUN_1010_0070	CompilerStrLen
FUN_1010_1774	CompilerStrCpy
FUN_1010_1a28	CompilerStrCat
FUN_1038_06f8	CompilerStrHelper

Nameable Functions (pattern)

Full list is in cmp_nameable.tsv (copied into this directory). Below is a sample of the first entries.

category	function
Memory	FUN_1008_1eea
Memory	FUN_1008_1f6c
Memory	FUN_1010_1030
Memory	FUN_1010_1352
Memory	FUN_1010_1898
Memory	FUN_1010_1b56
Memory	FUN_1040_0000
Memory	FUN_1040_0e1c
Memory	FUN_1040_0f08
Memory	FUN_1040_0fa6
Memory	FUN_1078_0400
Memory	FUN_1078_0530
Memory	FUN_1080_12ba
Memory	FUN_1088_15ba
Memory	FUN_10a8_004e
Memory	FUN_10a8_0130
Memory	FUN_10f8_0000
Message	FUN_1088_0fca
Property	FUN_1008_1848
Property	FUN_1010_0fda
Property	FUN_1028_0000
Property	FUN_1028_8ce8

category	function
Property	FUN_1038_0b24
Property	FUN_1048_18a8
Property	FUN_1070_0c60
Property	FUN_1078_0530
Property	FUN_1080_0b4c
Property	FUN_1088_15ba
Property	FUN_1088_1984
Property	FUN_10a8_12c4
Property	FUN_10a8_199a
Property	FUN_10c0_0c24
Property	FUN_10c0_0cb0
Property	FUN_10e0_0000
Property	FUN_10e0_00d4
Resource	FUN_1010_0f2a
Resource	FUN_1040_0f08
String	FUN_1008_2000
String	FUN_1010_0ee6
String	FUN_1010_0f2a

(showing first 40 of many; see TSV for complete list)

Notes

Rename script: rename_cmp.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80RTM Known Names

Main engine, windowing, page paint

Confirmed Renames

Original	Name
FUN_10e0_0050	PaintRectangle
FUN_1398_05c6	PaintRectangleAlt
FUN_1070_0b32	BitBltHelper
FUN_10d0_2d4e	BitBltHelper2
FUN_10d0_2fa0	BitBltHelper3
FUN_10e0_0dea	BitBltHelper4
FUN_1130_32ce	BitBltHelper5
FUN_1010_1a0a	SendMessageHelper

Nameable Functions (pattern)

Full list is in rtm_nameable.tsv (copied into this directory). Below is a sample of the first entries.

category	function
File	FUN_1008_485e
File	FUN_1090_048c
File	FUN_1150_0000
File	FUN_1190_11e8
File	FUN_1190_154c
File	FUN_1240_0888
File	FUN_1240_1232
File	FUN_1240_12b4
File	FUN_1240_13c2
File	FUN_1268_0882
GDI	FUN_1008_068c
GDI	FUN_1040_0338
GDI	FUN_1040_041a
GDI	FUN_1058_08bc
GDI	FUN_1058_098c
GDI	FUN_1068_0000
GDI	FUN_1068_0034
GDI	FUN_1070_0b32
GDI	FUN_1070_14d4
GDI	FUN_1090_0284
GDI	FUN_10b8_1588
GDI	FUN_10b8_17dc

category	function
GDI	FUN_10b8_26f0
GDI	FUN_10b8_3192
GDI	FUN_10c8_125a
GDI	FUN_10d0_0fd6
GDI	FUN_10d0_1d4a
GDI	FUN_10d0_2d4e
GDI	FUN_10d0_2fa0
GDI	FUN_10e0_0050
GDI	FUN_10e0_022a
GDI	FUN_10e0_0d5c
GDI	FUN_10e0_0dea
GDI	FUN_1130_2404
GDI	FUN_1130_32ce
GDI	FUN_1130_35b0
GDI	FUN_1130_366e
GDI	FUN_1130_38e8
GDI	FUN_1130_3934
GDI	FUN_1138_060e

(showing first 40 of many; see TSV for complete list)

Notes

Rename script: rename_rtm.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80UTL Known Names

Object type table and helpers

Confirmed Renames

Original	Name
FUN_1000_093b	UtlAlloc
FUN_1018_1472	UtlLock
FUN_10b8_0eac	UtlAlloc2
FUN_10c0_028c	UtlLock2
FUN_1008_0504	UtlStrCpy
FUN_10b8_0000	UtlStrLen
FUN_10c8_001e	UtlStrCat

Nameable Functions (pattern)

Full list is in utl_nameable.tsv (copied into this directory). Below is a sample of the first entries.

category	function
File	FUN_1030_0290
File	FUN_1030_03a4
File	FUN_1030_03f0
GDI	FUN_1060_01d6
Memory	FUN_1000_093b
Memory	FUN_1008_0504
Memory	FUN_1018_00c9
Memory	FUN_1018_0230
Memory	FUN_1018_06e0
Memory	FUN_1018_06f2
Memory	FUN_1018_074a
Memory	FUN_1018_1472
Memory	FUN_1018_1682
Memory	FUN_1018_24fc
Memory	FUN_1030_0290
Memory	FUN_1030_03f0
Memory	FUN_1038_0d32
Memory	FUN_1038_0d90
Memory	FUN_1038_0df8
Memory	FUN_1038_0f5c
Memory	FUN_1078_0000
Memory	FUN_1078_0c8c
Memory	FUN_1080_0122

category	function
Memory	FUN_1080_018e
Memory	FUN_1080_04fe
Memory	FUN_10a8_0520
Memory	FUN_10b8_0e90
Memory	FUN_10c0_030a
Message	FUN_1018_06e0
Message	FUN_1078_01b2
Message	FUN_1078_0274
Property	FUN_1018_06e0
Property	FUN_1018_126d
Property	FUN_1018_284e
Property	FUN_1088_04e2
Property	FUN_10c8_0000
Resource	FUN_1030_03a4
Resource	FUN_1038_0800
String	FUN_1000_093b
String	FUN_1008_0504

(showing first 40 of many; see TSV for complete list)

Notes

Rename script: rename_utl.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80W16 Known Names

Thunk layer and graphic IDs

Confirmed Renames

Original	Name
FUN_1010_1914	W16Alloc
FUN_1058_0f14	W16Lock
FUN_1058_2568	W16Alloc2
FUN_10b0_0060	W16Lock2
FUN_1030_003a	W16StrCpy
FUN_10b0_022c	W16StrLen
FUN_10c0_013a	W16StrCat
FUN_10c0_0310	W16StrHelper

Nameable Functions (pattern)

Full list is in w16_nameable.tsv (copied into this directory). Below is a sample of the first entries.

category	function
File	FUN_10b0_0144
File	FUN_10b0_2d90
File	FUN_10e8_01c0
GDI	FUN_1010_034c
GDI	FUN_10b0_0144
GDI	FUN_10e8_01c0
Memory	FUN_1010_034c
Memory	FUN_1010_035c
Memory	FUN_1010_1914
Memory	FUN_1010_1cde
Memory	FUN_1058_0de0
Memory	FUN_10a0_0ee2
Memory	FUN_10b0_008a
Memory	FUN_10b0_0144
Memory	FUN_10c0_0016
Memory	FUN_10c0_00d0
Memory	FUN_10f0_01ba
Message	FUN_1038_0992
Message	FUN_1058_0b2c
Message	FUN_1058_0de0
Message	FUN_10a0_0ee2
Property	FUN_1010_054e

category	function
Property	FUN_1038_04ee
Property	FUN_1070_0000
Property	FUN_10b0_0144
Property	FUN_10e8_0000
Property	FUN_10e8_01c0
Resource	FUN_1068_0544
String	FUN_1010_034c
String	FUN_1010_035c
String	FUN_1010_0632
String	FUN_1010_08c0
String	FUN_1010_10ad
String	FUN_1010_1154
String	FUN_1038_000c
String	FUN_1038_0992
String	FUN_1058_0de0
String	FUN_1068_0894
String	FUN_10a0_077c
String	FUN_10a0_0ee2

(showing first 40 of many; see TSV for complete list)

Notes

Rename script: rename_w16.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80LNL Known Names

LNL helpers

Confirmed Renames

Original	Name
FUN_1018_04a6	LnlAlloc
FUN_1050_043a	LnlLock
FUN_1070_203e	LnlAlloc2
FUN_10d8_097e	LnlLock2
FUN_1040_344c	LnlStrCpy
FUN_1040_40f0	LnlStrLen
FUN_1090_0130	LnlStrCat

Nameable Functions (pattern)

Full list is in lnl_nameable.tsv (copied into this directory). Below is a sample of the first entries.

category	function
File	FUN_1010_0094
File	FUN_1010_0408
File	FUN_1020_0084
File	FUN_1050_04c2
File	FUN_1090_092a
File	FUN_10d8_05fe
File	FUN_1100_043a
GDI	FUN_10a0_35d8
Memory	FUN_1010_0094
Memory	FUN_1018_051c
Memory	FUN_1018_05c0
Memory	FUN_1020_0744
Memory	FUN_1028_0440
Memory	FUN_1038_05e4
Memory	FUN_1048_0000
Memory	FUN_1050_0002
Memory	FUN_1050_04c2
Memory	FUN_1070_010e
Memory	FUN_1070_203e
Memory	FUN_1070_20c0
Memory	FUN_10d8_0000
Memory	FUN_10f8_170c
Memory	FUN_10f8_18c8

category	function
Memory	FUN_10f8_1cd2
Memory	FUN_1110_0000
Memory	FUN_1110_00cc
Message	FUN_1018_0096
Message	FUN_1020_012a
Message	FUN_1028_0050
Message	FUN_1028_00d0
Message	FUN_1030_0238
Message	FUN_1030_0ede
Message	FUN_1030_0f92
Message	FUN_1030_0fbe
Message	FUN_1030_10dc
Message	FUN_1040_00d4
Message	FUN_10a0_35d8
Message	FUN_10b0_0760
Message	FUN_10c8_396e
Message	FUN_10d8_19d2

(showing first 40 of many; see TSV for complete list)

Notes

Rename script: rename_lnl.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80MM Known Names

Multimedia notifications

Confirmed Renames

Original	Name
FUN_1070_03f4	MmSendMessage
FUN_1088_0200	MmSendMessage2
FUN_1088_0642	MmSendMessage3
FUN_1088_09f0	MmCreateWindow
FUN_1048_00d0	MmAlloc
FUN_1098_006a	MmAllocStr
FUN_1098_0134	MmStrCpy

Nameable Functions (pattern)

No nameable TSV available for this module.

Notes

Rename script: rename_mm.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80RCR Known Names

RCR graphics

Confirmed Renames

Original	Name
FUN_1008_158c	RcrBitBlt
FUN_1008_164c	RcrBitBlt2
FUN_1008_170e	RcrBitBlt3
FUN_1008_077a	RcrSendMessage
FUN_1008_0e94	RcrSendMessage2
FUN_1008_11aa	RcrSendMessage3
FUN_1008_08e2	RcrAlloc
FUN_1008_2190	RcrAlloc2
FUN_1008_aa62	RcrStrCpy

Nameable Functions (pattern)

No nameable TSV available for this module.

Notes

Rename script: rename_rcr.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80FLT Known Names

FLT graphics

Confirmed Renames

Original	Name
FUN_1028_14be	FltBitBlt
FUN_1040_071a	FltBitBlt2
FUN_1028_0390	FltSendMessage
FUN_1028_2316	FltSendMessage2
FUN_1028_26ac	FltSendMessage3
FUN_1010_0e98	FltAlloc
FUN_1010_20b6	FltAlloc2
FUN_1000_093b	FltStrCpy

Nameable Functions (pattern)

No nameable TSV available for this module.

Notes

Rename script: rename_flt.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.

TB80NET Known Names

Network layer

Confirmed Renames

Original	Name
FUN_1038_01b4	NetSendMessage
FUN_1038_0484	NetSendMessage2
FUN_1040_0000	NetSendMessage3
FUN_1008_17c8	NetCreateWindow
FUN_1008_0862	NetAlloc
FUN_1008_136c	NetAlloc2
FUN_1020_0410	NetStrCpy
FUN_1020_0a10	NetStrCpy2

Nameable Functions (pattern)

No nameable TSV available for this module.

Notes

Rename script: rename_net.py. Run from Ghidra Script Manager on the corresponding project. Confirmed names are the ones that have been manually verified against GDI calls, SoftICE traces, or clear property-ID patterns.